MATH 440/540 **HW 1 - ODE Review** Schreyer

Due Thurs. Aug. 22

WITH SOLUTIONS

These are practice problems to review your ODE skills for solving ordinary differential equations that will come up rather frequently in our PDE course (soon!). Please use these as a guide for what you may need to review further (this material is covered in any undergraduate ordinary differential equations textbook). It is expected that you would be able to solve these sorts of problems *quickly*. It is expected that you would (even after the class starts) need to look up the theorems on existence and uniqueness (so please have a good undergraduate ODE book handy!) but that you understand what the theorems mean after reading the theorem only.

The problems are given on the last two pages.

When turning in homework:

1. Summarize each problem statement (you can either write in your own words or write out the problems as you see written here or as they are written in the text). It should be clear what the goal of the problem is from your statement. At the end of the semester you should be able to look at your homework problems and understand what the goal of the problem is *without referring to external resources, including our textbook or e.g. this homework sheet.* **Note: I do take points off if you do not have a problem statement.**

2. Write your solution with sufficient explanation so that a classmate *who does not understand how to do the problem* can follow your work. You do not have to explain how you integrate or differentiate simple expressions, but you should explain why you are doing what you are doing. It is expected that if you picked up your homework 6 months from now you could understand what you are doing *without referring to external resources.* Again, I do take off points if there is insufficient explanation.

3. As always, follow the academic honor code. Remember that plagiarism is representing someone else's ideas as your own. That someone else could be a classmate, your instructor, another textbook, the internet, etc. If you use the internet to get an idea for how to do a problem, be sure to cite the webpage. If you see a classmates solution and you use that student's ideas, give credit to the student. *I do not take off points if you cite your sources. But if you plagiarize and are caught the consequences can be serious, either from a warning (first time only), to expulsion.*

4. To be successful in this class, you will need to know how to do these sorts of problems. If you do not understand how to do a problem, use your resources: classmates, email

your instructor (me), come see me during office hours, go to the MLC, etc. We will not have time to review this material during class time, so take the time and review the material *now*, as it will be assumed you will know this material for this class. If you are rusty on one (or likely more than one) of these problems, work several other problems of the same type.

1 Concepts

References: Jiri Lebl (*Notes on Diffy Qs, Differential Equations for Engineers*), available as pdf free online
Paul's notes are good for existence and uniqueness theorems.

1.1 A linear ordinary differential equation

is an equation that has the form

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}\frac{d^{n-1}y}{dx^{n-1}} + ... + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

Here we are looking for a solution of the form $y = f(x)$, where y is the **dependent variable** and x is the **independent variable**. Note that all coefficients are only a function of the independent variable. If $g(x) = 0$ we say the equation is **homogeneous**, and if $g(x) \neq 0$ we say the equation is **nonhomogeneous**.

Some ODE's need to be rearranged to determine whether it is a linear equation or not. For example:

$$\frac{1}{y}y' = \frac{1}{x} + \frac{x}{y}$$

may look like it is a nonlinear equation because the coefficient of y' is not just a function of the independent variable, x. However multiplying through by y gives $y' = \frac{y}{x} + x$ or $y' - \frac{1}{x}y = x$, and we see that $a_1(x) = 1$, $a_0(x) = -\frac{1}{x}$, and $g(x) = x$. Since we can write it in the above form, the equation is indeed linear.

Reference:
More examples for first-order linear vs nonlinear equations can be found:
https://www.myphysicslab.com/explain/classify-diff-eq-en.html

1.2 Existence and uniqueness for an initial value problem (IVP)

1.2.1 Existence and uniqueness for a linear first-order initial value problem (IVP)

Theorem:

Suppose we have the following IVP (note that this is a first-order *linear* ODE)

$$y'(x) + p(x)y = g(x), \qquad y(x_0) = y_0$$

Note that the coefficient of y' is 1. The solution is said to exist if we can find a *function*, $y = f(x)$ (when graphed must pass the vertical line test) that, (i) when substituted back into the equation, we get it is satisfied exactly for all x in an open interval containing x_0, and (ii) the function satisfies the initial condition, $y(x_0) = y_0$.

If we can find α and β such that

1. $\alpha < x_0 < \beta$
2. $p(x)$ and $g(x)$ are continuous on the open interval $\alpha < x < \beta$,

Then there is a unique solution to the IVP for all x, $\alpha < x < \beta$.

Notes:

1. If we cannot find such an interval, then we might have a unique solution or we might not. According to this theorem we don't know!
2. The bigger the interval, the better (the solution, a function, is well defined for a longer time), so we always give the biggest interval!

1.2.2 Existence and uniqueness for a nonlinear first-order initial value problem (IVP)

Theorem:

Suppose we have the following *nonlinear* IVP

$$y'(t) = g(t, y), \qquad y(t_0) = y_0$$

Note that the coefficient of y' is again 1. Again, the solution is said to exist if we can find a *function*, $y = f(t)$ that, when substituted back into the equation, we get it is satisfied exactly for all t in an open interval containing t_0, and (ii) the function satisfies the initial condition, $y(t_0) = y_0$.

If we can find a *rectangle*, R, $\alpha < t < \beta$, $\gamma < y < \delta$ such that

1. (t_0, y_0) is in the rectangle
2. $g(t, y)$ and $\frac{\partial g}{\partial y}$ are continuous on R,

Then there is a unique solution to the IVP for some interval, $t_0 - h < t < t_0 + h$ contained in (α, β).

Notes.

1. If we cannot find such a rectangle, R, then the solution may still exist and be unique, but based on this theorem we do not know.

2. The biggest difference between this theorem and the one for linear differential equation, is that in the nonlinear case, we do not know what h_0 is - it might be such that $t_0 - h = \alpha$ and $t_0 + h = \beta$, or it could be very very small! Solutions to nonlinear ODE's can have tricky behavior!

Reference:
More examples for existence and uniqueness for first-order DE:
https://tutorial.math.lamar.edu/classes/de/IoV.aspx
Jiri Lebl (*Notes on Diffy Qs*) Theorem 1.2.1

Existence and uniqueness for a *linear* second-order initial value problem (IVP) Theorem :

Suppose we have the following IVP

$$y''(x) + p(t)y' + q(t)y = g(t), \qquad y(t_0) = y_0, \quad y'(t_0) = y'_0$$

Note that the coefficient of y'' is 1. The solution is said to exist if we can find a *function*, $y = f(t)$ that, (i) when substituted back into the equation, we get it is satisfied exactly for all t in an open interval containing t_0, and (ii) it satisfies the initial conditions.

If we can find an open interval $I = (\alpha, \beta)$ such that

1. t_0 is in I
2. $p(t)$, $q(t)$, and $g(t)$ are continuous in I,

then there is a unique solution to the IVP for all t in I.

Notes:

1. Note that the initial condition for y and y' are at the same point, t_0. In practice, almost any 2 conditions will work, i.e. $y(0) = 1$, $y(1) = 5$; $y(1) = 3$, $y'(5) = -2$. The initial conditions need to be such that when you are solving for the two constants using the initial conditions, there is only one solution.

2. Because of this theorem, if you find a solution to the IVP (even by guess and check) you are done!

3. We can tell, even before we start solving the IVP, where the solution might have difficulties.

Reference:
References for existence and uniqueness for second-order DE:
http://www.ltcconline.net/greenl/courses/204/ConstantCoeff/uniquenessExistence.htm
Jiri Lebl (*Notes on Diffy Qs*) Theorem 2.3.2

HW 1

1. For each of the initial value problems below

   (i) (2 pts) Identify the dependent variable and the independent variable

   (ii) (2 pts) Determine whether the ODE is linear or nonlinear

   (iii) (2 pts) Determine the interval over which we are guaranteed a solution

   (iv) (4 pts) Find the *explicit* solution to the IVP

   (v) (2 pts) Identify the interval over which your solution exists (as a *function*), and note whether it is consistent with the existence and uniqueness theorem.

   Example 1 - do NOT turn in

$$\frac{du}{dv} = \frac{u+1}{v}, \qquad u(1) = 3$$

   Solution

   i. The dependent variable is u and the independent variable is v, so the solution is a function, $u(v)$.

   ii. The governing equation can be written as

$$\frac{du}{dv} - \frac{1}{v}u = -\frac{1}{v}$$

   so the equation is *linear*.

   iii. Using the existence and uniqueness theorem of linear first-order equations, we see that we need to find an interval, (α, β) where 1 is in the interval, and where $\frac{1}{v}$ is continuous. The function $\frac{1}{v}$ is continuous for $v < 0$ and $v > 0$. The only interval that contains 1 is $v > 0$. Therefore we are guaranteed a unique solution that exists for $v > 0$.

   iv. Using "separation of variables" we have

$$\begin{aligned}
&\frac{du}{dv} = \frac{u+1}{v} \\
&\frac{du}{u+1} = \frac{dv}{v} \\
&\ln(u+1) = \ln(v) + C \\
&e^{\ln(u+1))=e^{\ln(v)}+C} \\
&u + 1 = e^C e^{\ln(v)} = Cv \\
&u = Cv - 1
\end{aligned}$$

   Using the initial condition, $v = 1$, $u = 3$ we get $C = 4$. Thus **the particular solution is**

$$u = 4v - 1$$

Let us check our answer. We need to check that the solution satisfies the ODE and the initial condition (IC). The easy part is the IC: $u(1) = 4(1) - 1 = 3.$:). Now the ODE. We evaluate the left side of the equation and then the right side:

$$\begin{aligned} \mathrm{L}HS &= \frac{du}{dv} = 4 \\ \mathrm{R}HS &= \frac{u+1}{v} = \frac{4v-1+1)}{v} = 4. \end{aligned}$$

Since the left-hand side and right-hand side are equal, we know we have a solution!

v. This solution $u = 4v - 1$ exists and is a solution for all v, $-\infty < v < \infty$. Note that technically we have a difficulty at $v = 0$, because when we checked our solution we are dividing by v. However this singularity is *removable*. This is consistent with our existence and uniqueness theorem, in that the solution does exist for $v > 0$ (it also exists for $v \leq 0$, but the theorem does not say that $v \leq 0$ is the only place the solution exists).

Example 2 - do NOT turn in

$$\frac{du}{dv} = \frac{v+1}{u}, \qquad u(0) = -2.$$

i. The dependent variable is u and the independent variable is v, so the solution is a function, $u(v)$.

ii. The governing equation *cannot* be written as

$$\frac{du}{dv} + p(v)u = g(v)$$

so the equation is *nonlinear*.

iii. Using the existence and uniqueness theorem of nonlinear first-order equations, we see that we need to find a rectangle, R, such that the initial condition, $(0, -2)$, is in R and where $f(u, v) = \frac{v+1}{u}$ is continuous and $\frac{\partial f}{\partial u}$ is continuous (take the partial derivative with respect to the *dependent* variable). We note that

$$f(u, v) = \frac{v+1}{u} \qquad \frac{\partial f}{\partial u} = -\frac{v}{u^2}$$

are both continuous in the rectangles $R_1 : \ -10 < v < 10, \ -10 < u < 0$ and $R_2 : \ -10 < v < 10, \ 0 < u < 10$ (we could replace 10 with ∞, but the way our theorem is worded, then rectangles should be bounded. The point is that a rectangle exists, and any rectangle where 10 is replaced by a positive number works). The rectangle that

contains the initial condition $v = 0$ and $u = -2$ is R_1. Since we have found a rectangle that satisfies the conditions of the existence and uniqueness theorem, we are guaranteed that there is an open interval containing $v = 0$ where the solution exists.

iv. We can solve this nonlinear equation using separation of variables:

$$\begin{aligned} u\,du &= (v+1)dv \\ \frac{1}{2}u^2 &= \frac{1}{2}v^2 + v + C \\ u^2 &= v^2 + 2v + C \end{aligned}$$

Using the initial condition, $v = 0$, $u = -2$ we get $C = 4$, and **the implicit solution** is

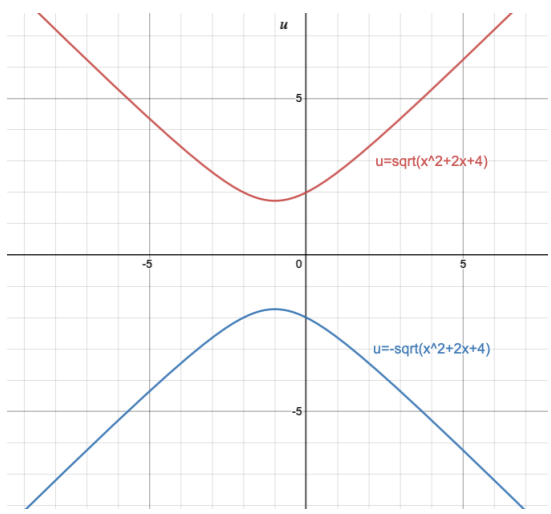
$$u^2 = v^2 + 2v + 4.$$

The solution should be a *function* so to find a function we take the square root:

$$u = \pm\sqrt{v^2 + 2v + 4}.$$

The graph of the implicit solution is given below. The implicit solution is a hyperbola (of the form $a^2 - b^2 = 1$ off-centered). This is *not* a function. We use the initial condition to determine that the solution is the negative branch:

$$u = -\sqrt{v^2 + 2v + 4}.$$



v. Writing the solution as a function (so it is single valued), we solve for u:

$$u(v) = \pm\sqrt{v^2 + 2v + 4}$$

This however is not a function (for each v we get two values of u!). Looking back at the solution from part (iv) we see this is a hyperbola!

Looking at the initial condition we see that we need the minus sign, so

$$u(v) = -\sqrt{v^2 + 2v + 4}$$

This is the lower half (in black) of the graph of the implicit solution. Where does this solution exist? It exists where $v^2 + 2v + 4 \geq 0$. Finding the zeros of $v^2 + 2v + 4$ (a parabola) we get $v = 1 \pm \sqrt{3}i$. So this parabola never crosses the v-axis - it is either always positive or always negative. Checking one point (all that is needed) we see that $v^2 + 2v + 4$ is always positive. Thus the interval containing the initial point $v = 0$ for which the solution exists is $-\infty < v < \infty$, i.e. the entire real line.
The existence of the solution is consistent with our answer to (iii), in that there is an open interval around $v = 0$ where the solution exists.

(a) (12 pts)

$$\frac{dy}{dx} = y^2, \qquad y(0) = 1$$

i. The dependent variable is y and the independent variable is x, so the solution is a function, $y(x)$.

ii. Because the coefficient in front of $\frac{dy}{dx}$ is one, and the right side is the dependent variable squared, the governing equation is cannot be written in the form

$$\frac{dy}{dx} + p(x)y = g(x)$$

so it is *nonlinear*.

iii. Using the existence and uniqueness theorem of nonlinear first-order equations, we see that we need to find a rectangle, R, such that the initial condition, $(0, 1)$, is in R and where $f(x, y) = y^2$ is continuous and $\frac{\partial f}{\partial y}$ is continuous (take the partial derivative with respect to the *dependent* variable). We note that

$$f(x, y) = y^2 \qquad \frac{\partial y}{\partial x} = 2y$$

are both continuous in the rectangles $R_1 : \ -10 < x < 10, \ -3 < y < 3$ and $R_2 : \ -5 < x < 5, \ -5 < y < 5$. Both rectangles work since they contains the initial condition $x = 0$ and $y = 1$. Since we have found a rectangle that satisfies the conditions of the existence and uniqueness theorem, we are guaranteed that there is an open interval containing $x = 0$ where the solution exists.

iv. We can solve this nonlinear equation using separation of variables:

$$\frac{dy}{y^2} = dx$$
$$\frac{-1}{y} = x + C$$

Using the initial condition, $x = 0$, $y = 1$ we get $C = -11$, and **the implicit solution** is

$$\frac{-1}{y} = x - 1$$

v. To get an explicity solution, we solve for y:

$$y = \frac{1}{1-x}$$

Where does this solution exist? It exists where $1 - x \neq 0$, or $x > 1$ or $x < 1$. The interval that contains the initial condition is $x < 1$. The existence of the solution is consistent with our answer to (iii), in that there is an open interval around $x = 0$ where the solution exists.

(b) (12 pts)

$$(t-3)x' = -2x, \qquad x(1) = 4$$

i. The dependent variable is x and the independent variable is t, so the solution is a function, $x(t)$.

ii. The governing equation *can* be written as

$$\frac{dx}{dt} + p(t)x = g(t)$$
$$\frac{dx}{dt} + \frac{2}{t-3}x = 0$$

so the equation is *linear*.

iii. Using the existence and uniqueness theorem of linear first-order equations, we see that we need to find an interval, (α, β) where 1 is in the interval, and where $\frac{2}{t-3}$ is continuous. The function $\frac{2}{t-3}$ is continuous for $t < 3$ and $t > 3$. The only interval that contains 1 is $t < 3$. Therefore we are guaranteed a unique solution that exists for $t < 3$.

iv. Using "separation of variables" we have

$$\frac{dx}{x} = \frac{-2\,dt}{t-3}$$
$$\ln|x| = -2\ln|t-3| + C$$
$$e^{\ln|x|} = e^{\ln|t-3|^{-2}} + C$$
$$x = C(t-3)^{-2}$$

Using the initial condition, $t = 1$, $x = 4$ we get $C = 16$. Thus **the particular solution is**

$$x = \frac{16}{(t-3)^2}$$

Let us check our answer. We need to check that the solution satisfies the ODE and the initial condition (IC). The easy part is the IC: $x(1) = \frac{16}{(-2)^2} = 4.$:). Now the ODE. We evaluate the left side of the equation and then the right side:

$$\begin{aligned} \mathrm{L}HS &= (t-3)x' = (t-3)(-32)(t-3)^{-3} = -32(t-3)^{-2} \\ \mathrm{R}HS &= -2x = (-2)\left(\frac{16}{(t-3)^2}\right) = -32(t-3)^{-2} \end{aligned}$$

Since the left-hand side and right-hand side are equal, we know we have a solution!

v. This solution $x = \frac{16}{(t-3)^2}$ exists and is a solution for all t, $t > 3$ or $t < 3$. The initial condition is at $t = 1$, so the solution to the IVP exists for all $t < 3$, which is consistent with what we found in part (c)!

2. (4 pts) Prove the principle of superposition: If y_1 and y_2 are solutions to

$$y'' + p(t)y' + q(t)y = 0$$

then $y = c_1y_1 + c_2y_2$ is also a solution.

Since y_1 and y_2 are solutions we have

$$y_1'' + p(t)y_1' + q(t)y_1 = 0 \qquad y_2'' + p(t)y_2' + q(t)y_2 = 0.$$

We need to show that $y = c_1y_1 + c_2y_2$ is a solution:

$$\begin{aligned} y'' + p(t)y' + q(t)y &= (c_1y_1'' + c_2y_2'') + p(t)(c_1y_1' + c_2y_2') + q(t)(c_1y_1 + c_2y_2) \\ &= c_1y_1'' + c_1p(t)y_1' + c_1q(t)y_1 + c_2y_2'' + c_2p(t)y_2' + c_2q(t)y_2 \\ &= c_1(y_1'' + p(t)y_1' + q(t)y_1) + c_2(y_2'' + p(t)y_2' + q(t)y_2) \\ &= c_1(0) + c_2(0) = 0. \end{aligned}$$

3. (2 pts) The principle of superposition, as stated above, applies to (circle one, no explanation necessary):
A) Linear second-order homogeneous differential equations
B) Nonlinear second-order homogeneous differential equations
C) Linear second-order non-homogeneous differential equations
D) Nonlinear second-order non-homogeneous differential equations.

A. Note that the above proof does not work for non-homogeneous or nonlinear equations.

4. (6 pts) Consider the following ODE:

$$x'' + 9x = 0.$$

Find the general solution.

Assume a solution of the form $x = e^{\lambda x}$. Since there is nothing in the equation indicating what the independent variable is, I will use t, but any other variable (besides x) is also perfectly correct. Then

$$\begin{aligned} x'' + 9x &= \lambda^2 e^{\lambda t} + 9e^{\lambda t} \\ &= e^{\lambda t}(\lambda^2 + 9) = 0. \end{aligned}$$

Since $e^{\lambda t}$ is never zero for finite t, the only way the above can hold is if

$$\lambda^2 + 9 = 0.$$

This is known as the *characteristic equation*. Solving for λ, we have $\lambda = \pm 3i$. Two complex solutions are:

$$e^{3it} \qquad e^{-3it}$$

We want two real solutions. Using Euler's formula, we have

$$e^{3it} = \cos(3t) + i\sin(3t)$$

The real part and the imaginary part of a complex solution are also a solution so we now have two *real* solutions:

$$\cos(3t) \qquad \sin(3t)$$

and the general solution is thus

$$y = C_1 \cos(3t) + C_2 \sin(3t)$$

Check the result by plugging this function back into the equation to see that we indeed get 0!

5. (10 pts) Find the general solution to the following nonhomogeneous equation by finding the general solution to the homogeneous equation and then finding the particular solution.

$$y'' + 8y' + 15y = e^{3t}.$$

We need to find the general solution to the homogeneous equation and then a particular solution. To find the homogeneous solution we again look for a solution of the form $y_h = e^{\lambda t}$ (here we know the independent variable is t by looking at the right-hand side). Then

$$\begin{aligned} y_h'' + 8y_h' + 15y_h &= \lambda^2 e^{\lambda t} + 8\lambda e^{\lambda t} + 15e^{\lambda t} \\ &= e^{\lambda t}(\lambda^2 + 8\lambda + 15) = 0. \end{aligned}$$

Since $e^{\lambda t}$ is never zero for finite t, the only way the above can hold is if

$$\lambda^2 + 8\lambda + 15 = 0 \qquad \Rightarrow (\lambda + 3)(\lambda + 5) = 0.$$

So we have two distinct roots and the general homogeneous solution is

$$y_h = C_1 e^{-3t} + C_2 e^{-5t}.$$

To find a particular solution we look at the form of the right-hand side and guess that a solution is of the same form: $y_p = Ae^{3t}$ (since all derivatives of exponential functions have the same form). Finding y_p' and y_p'' and substituting in we get

$$\begin{aligned} y_p'' + 8y_p' + 15y_p &= 9Ae^{3t} + 24Ae^{3t} + 15Ae^{3t} \\ &= e^{3t}(9A + 24A + 15A) = e^{3t}. \end{aligned}$$

In order for the above to be satisfied we need $48A = 1$ and we have $y_p = \frac{1}{48}e^{3t}$. Putting it all together we have

$$y = y_h(t) + y_p(t) = C_1 e^{-5t} + C_2 e^{-3t} + \frac{1}{48}e^{3t}$$

Check that when this function is put back into the original equation we do indeed get e^{3t}.

6. (16 pts) Consider the differential equation

$$y'' + 6y' + 9y = 0$$

(a) (6 pts) Find two linearly independent solutions. Be sure to show they are both solutions and that they are linearly independent (hint: Wronskian).

Assume a solution of the form $y = e^{\lambda t}$. Then

$$\begin{aligned} y'' + 2y' + 2y &= \lambda^2 e^{\lambda t} + 6\lambda e^{\lambda t} + 9e^{\lambda t} \\ &= e^{\lambda t}(\lambda^2 + 6\lambda + 9) = 0. \end{aligned}$$

Since $e^{\lambda t}$ is never zero for finite t, the only way the above can hold is if

$$\lambda^2 + 6\lambda + 9 = 0 \qquad \Rightarrow \qquad \lambda = -3$$

This is a real repeated root and the general solution is of the form

$$y = C_1 e^{-3t} + C_2 t e^{-3t}$$

Two linearly independent solutions are $y_1(t) = e^{-3t}$ and $y_2(t) = te^{-3t}$. They are solutions because they each satisfy the equation $y''+6y'+9y = 0$ and they are linearly independent as is shown by either calculating the Wronskian at $t = 0$ and showing it is not zero:

$$\begin{aligned} W(t) = \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix}_{t=0} &= \begin{vmatrix} e^{-3t} & te^{-3t} \\ -3e^{-3t} & e^{-3t} - 3te^{-3t} \end{vmatrix}_{t=0} \\ &= \begin{vmatrix} 1 & 0 \\ -3 & 1 \end{vmatrix} = 1 \neq 0. \end{aligned}$$

(Note: we have a theorem which says the Wronskian is either always zero or never zero, so it is sufficient to calculate the Wronskian at one point, here we chose $t = 0$).

or by noting that the only solution to $C_1 y_1(t) + C_2 y_2(t) = 0$ is $C_1 = C_2 = 0$ ($y_1(t)$ is not a scaler times $y_2(t)$).

(b) (2 pts) Write down the general solution.

$$y = C_1 e^{-3t} + C_2 t e^{-3t}$$

(c) (8 pts) Consider the initial value problem

$$y'' + 6y' + 9y = 0 \qquad y(0) = 5, \quad y'(0) = -10.$$

Give a physical interpretation of the problem and then find the solution.

Consider a mass, spring, dashpod system. The mass is 1kg, the damping coefficient is 6 kg/s, and the spring constant is 9 kg/s^2 or 9 N/m. Let y be the displacement of the mass from equilibrium. Start the mass is displaced 5m and give it an initial velocity of -10 m/s. The solution of this equation gives the motion of the mass for $t > 0$.
We use the initial conditions to determine the coefficients C_1 and C_2. The initial conditions give us:

$$\begin{aligned} y(0) &= C_1 = 5 \\ y'(0) &= -3C_1 e^{-3t} + C_2 e^{-3t} - 3C_2 t e^{-3t}\Big|_{t=0} = -3C_1 + C_2 = -10 \end{aligned}$$

Solving we get $C_1 = 5$ and $C_2 = 5$ and so:

$$y(t) = 5e^{-3t} + 5te^{-3t}$$

Checking at least the initial condition $y(0) = 5$ is satisfied we have hope to believe this is the correct solution without checking that this function satisfies the equation as well.